

PYTHON CONVERSATIONAL LOOP

30-Minute Student Task Report

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Task	Rule-Based Conversational Chatbot

1. Task Overview

The task is to build a simple rule-based conversational chatbot using Python. The chatbot should interact continuously with the user, understand different types of messages, and provide suitable responses based on predefined rules. It should first ask for the user's name, recognize greetings and different questions or topics, handle unknown messages with a useful response, and allow the user to end the conversation by typing **bye** or **exit**. The chatbot should use basic Python concepts such as **input()**, **.lower().strip()**, **while True**, **if/elif/else**, **in**, and **break**. The overall flow of the chatbot is **Input** → **Process** → **Respond** → **Repeat** → **Exit**.

2. Features Implemented

- Asks the user's name before the main conversation.
- Uses **while True** to keep the conversation running.
- Uses **input()** and cleans the message with **lower().strip()**.
- Recognizes greetings including hello, hi, and hey.
- Answers questions about AI, college, and subjects.
- Uses **if / elif / else** for decision making.
- Uses **break** when the user types bye or exit.
- Uses the user's name in chatbot responses.
- Handles unknown messages with a useful reply.
- Uses the **in** operator so greeting variations can be recognized.

3. Chatbot Code

```
# python

# print("=====")
# print("      ■ PYTHON CHATBOT")
# print("=====")

# Ask the user's name
name = input("Enter your name: ").strip()

print(f"\nHello {name}! ■")

print("I am your simple Python chatbot.")
print("You can ask me about Python, college, or technology.")
print("Type 'bye' or 'exit' to end the conversation.\n")

# Keep the conversation running
while True:
    user_input = input(f"{name}: ").lower().strip()

    # Exit condition
    if user_input == "bye" or user_input == "exit":
        print("Bot: Goodbye", name + "! Have a great day! ■")
        break

    # Greetings
    elif "hello" in user_input or "hi" in user_input:
        print(f"Bot: Hello {name}! How can I help you?")

    # Python topic
    elif "python" in user_input:
        print("Bot: Python is a popular programming language known for its simple syntax.")

    # College topic
    elif "college" in user_input:
        print("Bot: College is a great place to learn, build projects, and develop your career.")

    # Technology topic
    elif "technology" in user_input or "tech" in user_input:
        print("Bot: Technology helps us solve problems and build useful applications.")

    # Help
    elif "help" in user_input:
        print("Bot: You can ask me about Python, college, or technology.")

    # Unknown message
    else:
        print("Bot: Sorry, I don't understand that yet. Try asking about Python, college, or technology.")
```

4. Sample Execution

Enter your name: hari

Hello hari! ■

I am your simple Python chatbot.

You can ask me about Python, college, or technology.

Type 'bye' or 'exit' to end the conversation.

hari: What is python

Bot: Python is a popular programming language known for its simple syntax.

hari: what is ai

Bot: Sorry, I don't understand that yet. Try asking about Python, college, or technology.

hari: what are the technology are available

Bot: Technology helps us solve problems and build useful applications.

hari: tell me about college

Bot: College is a great place to learn, build projects, and develop your career.

hari: bye

Bot: Goodbye hari! Have a great day! ■

5. Result

The rule-based chatbot was successfully implemented and tested with greetings, different topics, an unknown message, and an exit command. The chatbot continues the conversation until the user enters **bye** or **exit**.